Voyance — Evaluation & Drift Report

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CONTENTS

Voyance — Evaluation & Drift Report 3

1. Summary 3

2. Benchmark (same split, same features) 3

3. Error decomposition (P50, test month) 4

By sea region 4

Feature importance (gain, top 12) 5

4. Drift monitoring 5

5. Method 6

6. Limitations 6

7. Provenance 6

Voyance — Evaluation & Drift Report

Generated: 2026-10-23 21:48 UTC · **Model:** xgboost-cuda-quantile-eta v1.2.0 · **Hardware:** NVIDIA GeForce RTX 4060 Laptop GPU (CUDA, tree_method=hist, device=cuda) **Scope:** cargo + tanker + passenger vessels (NOAA AIS)

1. Summary

Voyance is an open-source maritime voyage-intelligence platform grown from idealab idea #586938, pivoted per the owner to ship the model itself. This report is generated directly from pipeline artifacts; no number is hand-entered.

- Training months: 2024-01, 2024-04 · Test month: 2024-07
- Rows: 10,366 train / 8,227 test (voyage-disjoint CQR holdout inside train)
- P50 test MAE: **4.49 h** vs speed-prior baseline 5.35 h
- P10–P90 empirical coverage: **83.0%** (nominal 80%) after a conformal adjustment of ±2.73 h (CQR, Romano et al. 2019)

2. Benchmark (same split, same features)

Model	Hardware	Test MAE	MAPE
XGBoost quantile (served)	RTX 4060 GPU · CUDA	4.49 h	40.9%
LightGBM quantile	14-thread CPU	5.80 h	63.4%
Speed-prior baseline	—	5.35 h	50.6%

3. Error decomposition (P50, test month)

BY SEA REGION

Region	MAE (h)	Samples
us east coast	5.94	3,307
great lakes	1.80	2,212
us west coast	3.32	1,599
gulf of mexico	9.07	722
puerto rico caribbean	2.55	276
other us waters	7.99	90
hawaii pacific	1.09	21

Full breakdowns — Beaufort wind class, vessel type group, distance-to-go bucket — are in `reports/eval.json` and rendered live on the dashboard.

FEATURE IMPORTANCE (GAIN, TOP 12)

Feature	Gain
hours_at_type_speed	579.7
remaining_distance_nm	487.7
hours_at_current_speed	441.1
hours_at_sog6	172.3
arr_port_idx	119.7
dep_port_idx	96.3
wind_gusts_10m	68.5
precipitation	62.0
sin_wave_dir	54.9
wave_height	53.9
vessel_draft	53.1
speed_kn	51.1

4. Drift monitoring

Baseline feature distribution: 2024-01 (feature distribution reference). Reference MAE: 4.49 h. Alert rule: any feature PSI > 0.2 or MAE ratio > 1.25.

Slice	Rows	MAE (h)	MAE ratio	Drifted features (PSI)	Alert
2024-04	6,868	5.98	1.33	—	ALERT
2024-07	8,227	4.49	—	—	ok

HTML reports: `reports/drift/` (Evidently). Machine-readable alerts: `GET /v1/drift`.

5. Method

Voyages are segmented from NOAA MarineCadastre daily AIS broadcast points by 15 km UN/LOCODE port buffers (20 km clustering of adjacent gazetteer entries), with receiver-noise plausibility filters (US-waters bounding box, per-fix implied speed < 90 kn, per-voyage implied speed < 40 kn). Samples ≥ 1 h apart and ≥ 8 NM from the destination join hourly ERA5 reanalysis (wind, gusts, precipitation, temperature, humidity) and ERA5 wave data (height, period, direction, swell) on a 0.5° grid via Open-Meteo (CC BY 4.0). The target is remaining time to arrival at the destination port. Temporal split: train Jan+Apr 2024, test Jul 2024.

6. Limitations

US-waters AIS only; voyages cannot span the downloaded week boundaries; ERA5 smooths local weather; port queues are partially visible; tug/fishing vessels excluded by design. Not a certified navigation aid. Full list: `reports/model-card.md`.

7. Provenance

NOAA MarineCadastre AIS (CC0) · Open-Meteo ERA5 (CC BY 4.0) · UN/LOCODE (UNECE) · LightGBM/XGBoost/Evidently (MIT/Apache-2.0). Attribution: `docs/ATTRIBUTION.md`.